

- 8** An astronaut of mass  $m$  in a spacecraft experiences a gravitational force  $F = mg$  when stationary on the launchpad.

What is the gravitational force on the astronaut when the spacecraft is launched vertically upwards with an acceleration of  $0.2g$ ?

- A**  $1.2mg$                       **B**  $mg$                       **C**  $0.8mg$                       **D**  $0$

- 9** The diagram shows a child X of mass  $20\text{ kg}$  and a child Y of mass  $15\text{ kg}$  seated on a uniform